B.Tech (1st Semester) **GROUP-A** **Batch (2026-2030)**

Course Code -100102 **Engineering Mathematics – I** **3 0 0 3**

Course Outcomes (COs)	Course Outcome Statements
CO1	Apply the concept of Linear/ Matrix algebra to solve given engineering problems.
CO2	Demonstrate the ability to solve engineering problems based on the application of derivatives.
CO3	Solve the given optimization problems by using the concept of partial derivatives.
CO4	Evaluate the given multiple integrals in different coordinate systems to solve complex engineering problems.
CO5	Apply the concept of vector calculus to solve the given engineering problems.

Mapped CO(s) Number	Relevant Major Theory Session Outcomes (TSOs)	Relevant Units/ Body of Knowledge (BoK)
CO1	*TSO 1a.* Find the rank of a given Matrix by using various methods. *TSO 1b.* Test the Consistency of a given system of simultaneous linear equations representing an engineering situation. *TSO 1c.* Apply Cayley-Hamilton Theorem to find the integral powers of a given matrix. *TSO 1d.* Test the nature of the transformed matrix of a given homogeneous quadratic polynomial.	**Unit- 1.0: Basic Linear Algebra (8hrs)** 1.1 Elementary transformations and Rank of Matrix, Rank by using Echelon and Normal form of Matrices. Linear dependence & independence of vectors in R^n space 1.2 Consistency of System of Linear Equations 1.3 Eigenvalues & Eigenvectors, Cayley-Hamilton Theorem. 1.4 Similarity of Matrices & Diagonalization of Square Matrices. Quadratic Form, Canonical Form of Matrices.
CO2	*TSO 2a.* Find the nth order derivatives of given functions representing an engineering situation. *TSO 2b.* Apply the Mean Value Theorem to determine the nature of the tangent of given curves under certain conditions. *TSO 2c.* Evaluate limits of given functions applying L' Hopital Rule.	**Unit- 2.0: Differential Calculus: Single Variable (8hrs)** 2.1 Higher order derivatives; Successive differentiation and Leibnitz's Theorem 2.2 Mean Value Theorems (Rolle's; Lagrange's; Cauchy) 2.3 Indeterminate Forms; L' Hopital Rule 2.4 Tangents & Normal of Algebraic & Polar Curves, Curvature

Mapped CO(s) Number	**Relevant Major Theory Session Outcomes (TSOs)**	**Relevant Units/ Body of Knowledge (BoK)**
	TSO 2d. Find the Curvature of a given curve representing an engineering situation.	
CO3	*TSO 3a.* Apply Euler's theorem on homogeneous functions of m independent variables of degree n. *TSO 3b.* Test the differentiability of a given function of several variables. *TSO 3c.* Apply Taylor's Theorem to expand a function representing an engineering situation in the nhd. of a given point. *TSO 3d.* Optimize a given function by applying Lagrange's Multiplier Method for given engineering problems.	**Unit- 3.0: Differential Calculus: Several Variables (9hrs)** 3.1 Partial differentiations; Euler's theorems 3.2 Limit, Continuity and differentiability of functions of Several Variables 3.3 Taylor's theorem with various forms of remainders, Maclaurin's theorem and related problems 3.4 Maxima & Minima for functions of several variables; Lagrange's Multiplier Method
CO4	*TSO 4a.* Evaluate the given definite integral using Leibnitz's rule. *TSO 4b.* Find the area of the given plane figure by using double integration. *TSO 4c.* Find the Volume of the given 3-D figure by using triple integration. *TSO 4d.* Evaluate given improper integrals by using Beta & Gamma functions.	**Unit- 4.0: Integral Calculus (9hrs)** 4.1 Differentiation under the integral sign; Leibnitz's rule 4.2 Double integral; Change the order of integration; Evaluation of double integration by changing it into polar co-ordinates 4.3 Triple Integral; Evaluation of Triple integral by changing to Spherical Polar co-ordinates & Cylindrical polar co-ordinates 4.4 Beta; Gamma; Elliptic integral; Relation between Beta & Gamma functions, General Properties and related problems.
CO5	*TSO 5a.* Find the divergence and curl of a given vector point function representing a physical quantity *TSO 5b.* Apply Stoke's theorem to convert given line integral into surface integral. *TSO 5c.* Apply Gauss' theorem to convert given surface integral into volume integral.	**Unit- 5.0: Vector Calculus (8hrs)** 5.1 Scalar and vector valued functions; Concepts of gradient, divergence and curl and their geometrical and physical significance; tangent plane; directional derivative; 5.2 Line, surface and volume integrals - Statement of Green's, Stokes' and Gauss divergence theorems -verification and evaluation of vector integrals using them.

Reference/Books:

S. No.	Titles	Author(s)	Publisher and Edition with ISBN
1	Higher Engineering Mathematics	B. S. Grewal	Khanna Publishers, 44 Ed. ISBN No:9788174091154
2	Higher Engineering Mathematics	B. V. Ramana	Tata McGraw Hill New Delhi, 11th Ed, 2010, ISBN-13-978-0070634190
3	Advanced Engineering Mathematics	Erwing Kreyszig	John Wiley & Sons, 10th Ed, ISBN-978-0470-15836-5
4	A Textbook of Engineering Mathematics	N. P. Bali & Dr. Manish Goyal	University Science Press, 13th Ed, ISBN-978-93-83828-63-0
5	Understanding Engineering Mathematics	John Bird	Routledge; First Edition ISBN 9780415662840